

GUETTINGEN, Switzerland

WMO: 066210

Lat: 47.6017N

Lon: 9.2794E

Elev: 440

StdP: 96.15

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.6	(c) -5.9	(d) -13.1	(e) 1.3	(f) -7.4	(g) -9.4	(h) 1.8	(i) -4.6	(j) 10.1	(k) 7.6	(l) 8.9	(m) 6.8	(n) 3.1	(o) 50	(p) 0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.1	(c) 29.8	(d) 20.4	(e) 28.0	(f) 19.9	(g) 26.3	(h) 19.2	(i) 21.6	(j) 27.7	(k) 20.7	(l) 26.3	(m) 19.9	(n) 24.9	(o) 2.0	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	15.2	24.2	18.8	14.4	23.3	18.1	13.7	22.2	65.0	27.9	61.8	26.5	58.7	25.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	6.5	5.5	-9.9	32.8	2.7	1.3	-11.8	33.7	-13.4	34.5	-15.0	35.2	-17.0	36.1		
			-10.6	23.4	2.7	0.9	-12.5	24.0	-14.0	24.5	-15.5	25.0	-17.5	25.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	1.2	1.7	5.8	10.1	14.0	18.1	19.6	19.1	15.0	10.6	5.4	1.9
	DBStd	7.55	4.05	4.09	3.88	3.73	3.64	3.58	3.02	3.04	3.12	3.44	3.39	3.62
	HDD10.0	1126	274	233	136	44	7	1	0	0	2	36	142	252
	HDD18.3	3116	532	466	388	246	141	48	22	28	107	241	388	510
	CDD10.0	1219	1	0	6	48	131	244	297	282	151	54	4	1
	CDD18.3	167	0	0	0	0	7	41	61	51	7	0	0	0
	CDH23.3	1243	0	0	0	7	67	329	468	341	32	0	0	0
	CDH26.7	313	0	0	0	1	8	83	132	88	1	0	0	0
Wind	WSAvg	2.2	2.5	2.5	2.5	2.2	2.3	2.2	2.1	2.0	2.0	1.9	2.2	2.4
Precipitation	PrecAvg	1081	57	61	67	81	114	131	125	117	90	77	84	77
	PrecMax	1334	138	136	188	159	238	209	266	198	173	135	169	155
	PrecMin	801	6	19	23	25	24	49	54	39	42	10	9	5
	PrecStd	126	34	30	33	35	50	42	48	44	35	34	39	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	13.7	18.5	24.1	27.7	31.6	32.4	32.0	26.2	20.9	16.4	12.2
		MCWB	9.4	8.6	11.0	14.6	18.4	20.9	21.1	21.0	19.2	15.5	12.1	9.3
	2%	DB	10.9	11.0	15.8	21.6	25.2	29.1	29.9	29.0	24.1	18.6	13.6	10.2
		MCWB	8.2	7.4	9.8	13.1	16.9	20.2	20.4	20.6	18.4	14.9	11.0	8.4
	5%	DB	8.8	9.2	13.8	19.5	22.9	27.1	28.0	26.9	22.0	16.8	11.6	8.4
		MCWB	6.8	6.4	9.1	12.4	15.7	19.5	19.9	20.0	17.5	14.2	9.8	6.9
	10%	DB	6.8	7.3	11.9	17.1	20.8	25.0	26.0	25.0	20.1	15.3	10.2	6.9
		MCWB	5.3	5.1	8.2	11.4	14.8	18.6	19.3	19.2	16.5	13.4	8.8	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	9.6	11.9	15.5	19.5	22.4	22.9	22.6	20.1	17.0	12.5	10.3
		MCDB	12.6	13.0	16.5	22.3	25.7	29.3	30.1	28.6	24.5	19.4	15.0	11.1
	2%	WB	8.5	7.9	10.5	14.0	17.9	21.1	21.6	21.5	19.0	15.7	11.3	8.6
		MCDB	10.3	10.5	14.5	20.3	23.2	27.4	27.6	27.3	22.8	17.6	13.2	10.0
	5%	WB	6.8	6.6	9.5	12.8	16.6	20.0	20.6	20.6	18.0	14.6	10.2	7.2
		MCDB	8.5	8.9	13.0	18.2	21.4	25.7	26.2	25.6	21.2	16.3	11.5	8.3
	10%	WB	5.4	5.2	8.5	11.7	15.4	19.0	19.7	19.7	16.9	13.6	9.0	5.9
		MCDB	6.8	7.1	11.7	16.6	19.8	23.9	24.9	24.0	19.4	15.0	10.0	6.8
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.8	8.5	10.1	9.7	9.9	10.1	9.6	8.4	6.7	5.0	4.3
		MCDBR	7.3	9.2	12.2	14.4	13.7	13.5	13.5	12.7	10.8	8.8	7.8	6.3
		MCWBR	5.3	5.9	7.0	7.3	6.4	5.9	5.8	5.6	5.7	5.6	5.3	4.8
	5% WB	MCDBR	6.7	8.7	11.0	13.1	12.3	12.2	12.4	11.8	10.1	8.3	7.4	5.9
		MCWBR	5.1	5.9	6.6	6.9	6.0	5.6	5.6	5.5	5.5	5.6	5.3	4.8
Clear-Sky Solar Irradiance	taub	0.292	0.317	0.354	0.391	0.404	0.411	0.412	0.405	0.378	0.348	0.317	0.288	
	taud	2.436	2.368	2.342	2.272	2.277	2.298	2.287	2.325	2.380	2.450	2.484	2.446	
	Ebn at Noon	795	844	863	860	860	855	847	837	827	799	758	754	
	Edh at Noon	66	87	105	124	128	127	126	117	101	81	64	58	
All-Sky Solar Radiation	RadAvg	1.12	1.94	3.15	4.31	5.00	5.73	5.51	4.81	3.60	2.25	1.24	0.92	
	RadStd	0.12	0.27	0.42	0.61	0.51	0.53	0.50	0.50	0.37	0.24	0.15	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (110 neighbors)	N/A	N/A	N/A	N/A	+0.44	+0.37	N/A	N/A	+54	+15

Nomenclature: See separate page